

Predicting Gen-Z Social Media Addiction Levels Using Machine Learning

Shruthi Srinivasan

ITSC 3156 Professor Hieu Le

1. Introduction

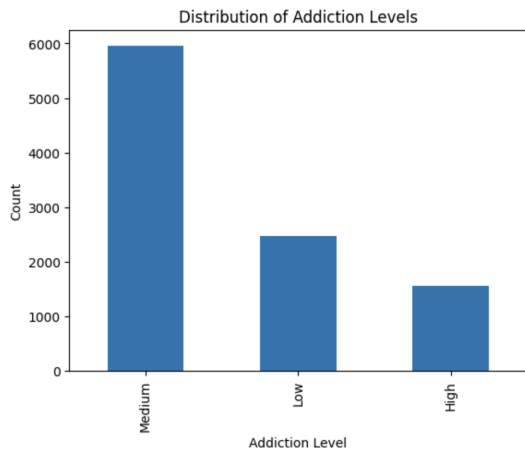
The use of social media has become such a normalized and significant part of our daily lives as Gen Z. Platforms such as Tiktok, Instagram and Snapchat provide vast opportunities for communication, entertainment and education, but excessive usage could lead to unhealthy behavioral patterns and signs of attachment to it. We can help bring awareness to the negative effects of social media by identifying the correlation between behavioral factors and digital well-being. The goal of this project is to predict the level of social media addiction among Gen-Z users using machine learning classification techniques. Addiction level prediction is approached as a supervised learning problem using behavioral features like daily usage hours, average session duration, number of platforms used and screen time before sleep to categorizing users into Low, Medium and High addiction levels.

Predicting addiction levels can help provide some insight on which usage habits are most related to the increased dependency. This can also help support future research related to digital well-being and behavioral analytics. Another motivation for this project was to apply classification techniques which we learned throughout the semester to a real-world problem. Through this project, we have the chance to practice preprocessing methods and evaluating model performance on something that we are passionate about.

In this project, I followed the standard machine learning workflow that included data exploration, preprocessing, model training, and evaluation. I first analyzed the dataset using visualizations to understand how features like daily usage hours, mental health score, and screen time before sleep relate to addiction level. After that, the categorical variables were converted into numerical values and feature scaling was applied to prepare the data for the models. The dataset was then split into training and testing sets using an 80/20 ratio. Finally, Logistic Regression and K-Nearest Neighbors were implemented and compared using accuracy scores, confusion matrices, and classification reports to determine which model performed better at predicting addiction level.

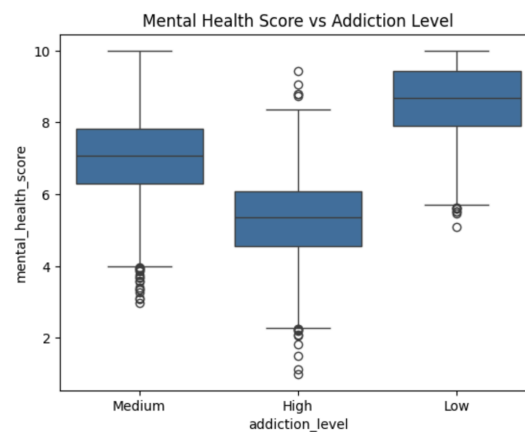
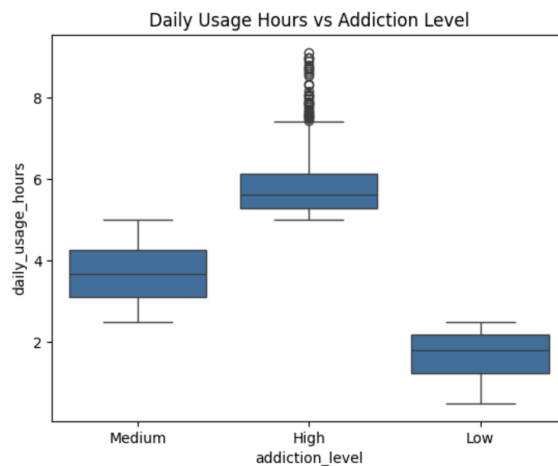
2. Data

The dataset used in this project is the Gen-Z Social Media Usage Dataset which I found in Kaggle. It includes information about social media usage behavior amongst only Gen-Z users from different countries. Age, gender, country, daily usage hours, number of platforms used, purpose of usage, average session duration, mental health score, night usage and screen time before sleep are some features that were useful for my prediction. The target variable in this dataset is addiction level, which categorizes users into Low, Medium or High levels of dependency. Since the original dataset had a massive number of entries, I selected a sample of 10,000 rows for this project. This helped improve efficiency in my code but also kept enough data to train and evaluate the machine learning models effectively.

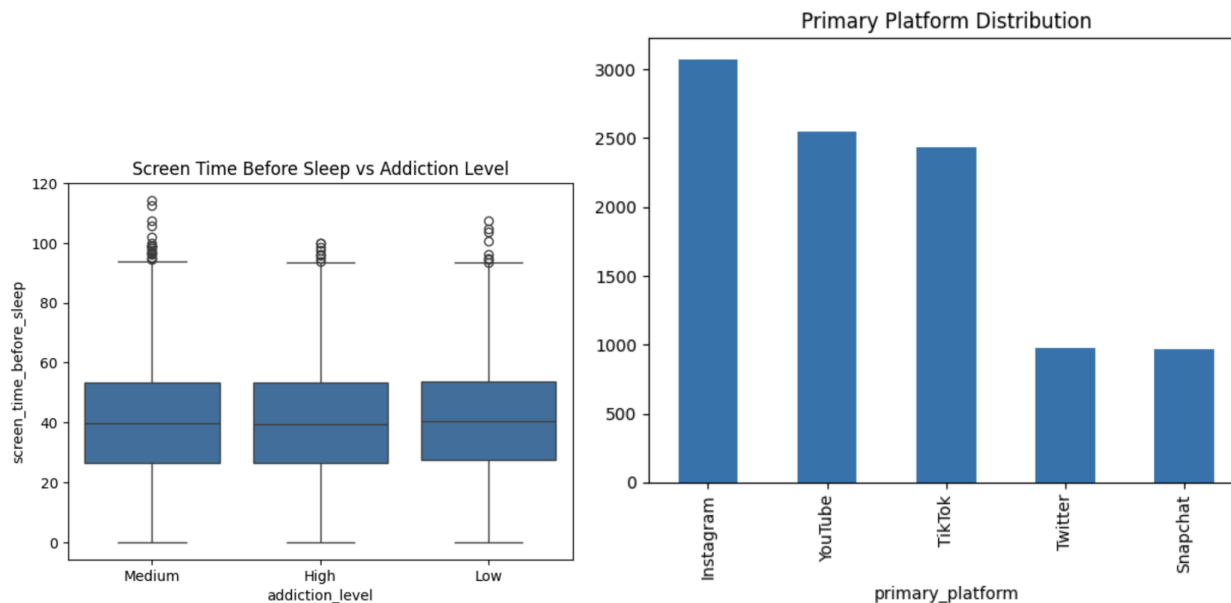


Before building the models, I explored the dataset using several visualizations to better understand how different features relate to addiction level. A bar chart showing the distribution of addiction levels confirmed that the dataset is fairly balanced across the Low, Medium, and High categories, which helps prevent the models from becoming biased toward one group. I also created boxplots comparing daily usage hours, screen time before sleep, and mental health score with addiction level. From these visualizations, it was clear that users with higher addiction levels generally spent more time on social media each day and also

tended to use social media more before going to sleep. Mental health score also showed noticeable differences between the groups, suggesting that it may play a role in predicting addiction level. I also looked at the distribution of primary platforms used to better understand overall usage patterns in the dataset.



Before training the models, I applied several preprocessing steps to prepare the dataset. Since machine learning models require numerical input, categorical features such as gender, country, primary platform, and purpose of usage were converted into numerical values using one-hot encoding. The addiction level column was also converted into numerical labels so that it could be used as the target variable for classification. After encoding the categorical variables, I split the dataset into training and testing sets using an 80/20 ratio. This resulted in 8,000 samples for training and 2,000 samples for testing. I also applied feature scaling to standardize the data so that all features were on a similar range. This step was especially important for the K-Nearest Neighbors model because it relies on distance between feature values during prediction.



3.1 Logistic Regression

The first machine learning algorithm that I used was Logistic Regression which is considered as the default one for classification purposes. This algorithm works by estimating the probability of the data point belonging to a particular class depending on the features of the dataset. In this study, it was applied to classify the level of the addiction of the users in terms of Low, Medium, or High according to the usage parameters like hours of using applications per day, the number of platforms utilized by the user, minutes of using the devices before sleeping, and the mental health of the individual. I chose this algorithm due to its good performance when the connection between the features and the target variable is apparent. As all of the features had evident correlations, this algorithm provided a reliable basis for further predictions.

3.2. K- Nearest Neighbors

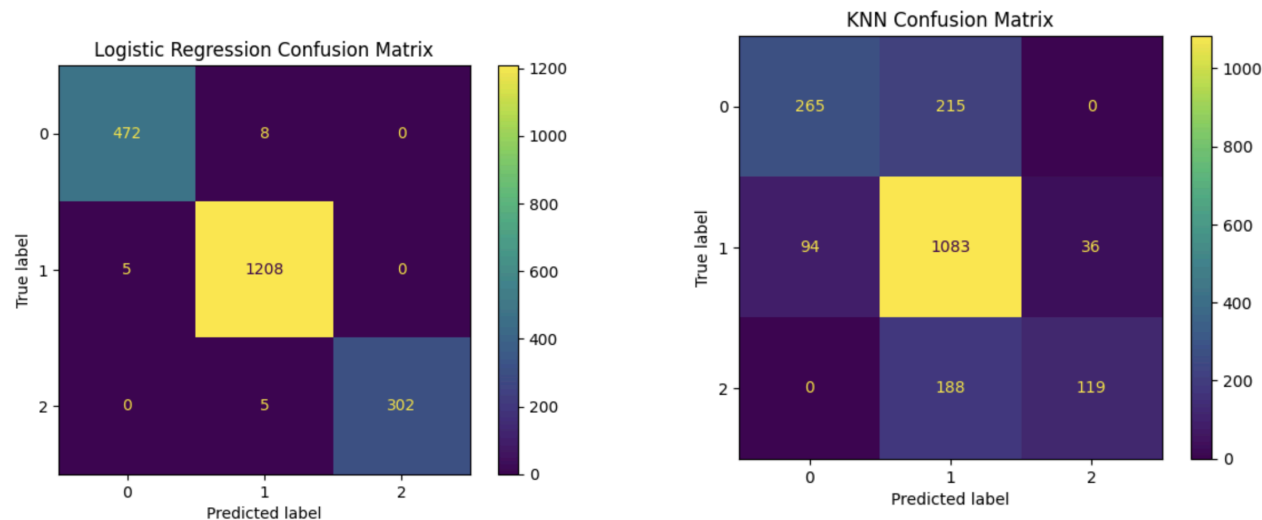
For the second model, I used K-Nearest Neighbors (KNN) to compare its performance with Logistic Regression. KNN is a classification algorithm that predicts the class of a data point by looking at the labels of nearby data points in the dataset. Instead of learning a formula like Logistic Regression, it makes predictions based on similarity between samples. Since KNN relies on distance between feature values, feature scaling was especially important before training this model. I selected a value of $k = 5$ after testing several values and observing how accuracy changed. Using KNN allowed me to compare how a distance-based classification method performs compared to a linear classification method on the same dataset.

4. Results

After preprocessing the dataset, I split the data into training and testing sets using an 80/20 ratio. This resulted in 8,000 samples used for training and 2,000 samples used for testing. Both Logistic Regression and K-Nearest Neighbors were trained on the training dataset and then evaluated using the testing dataset. To

measure model performance, I used accuracy scores, confusion matrices, and classification reports so I could compare how well each model predicted addiction levels across the three categories.

The Logistic Regression model achieved an accuracy of about 99% on the testing dataset. Based on the confusion matrix and classification report, the model correctly classified most samples across all three addiction levels with very few errors. Precision, recall, and F1-scores were also consistently high for each category, which shows that the model performed reliably across the dataset. These results suggest that Logistic Regression was able to capture clear relationships between behavioral features such as daily usage hours, screen time before sleep, and mental health score and the predicted addiction level. Since these features showed noticeable patterns during the visualization stage, it makes sense that a linear classification model performed well on this dataset.



The K-Nearest Neighbors model achieved an accuracy of about 73.35% on the testing dataset. Compared to Logistic Regression, the confusion matrix showed more misclassifications between the Medium and High addiction categories. The classification report also showed lower recall values for the High addiction level, meaning the model had more difficulty correctly identifying those users. Because KNN makes predictions based on similarity between nearby data points instead of learning relationships between features, it was less effective for this dataset. Even though feature scaling was applied before training the model, the overlapping behavioral patterns between addiction levels made it harder for KNN to separate the categories as accurately as Logistic Regression.

When comparing the two models, Logistic Regression clearly performed better than K-Nearest Neighbors for predicting addiction level. It achieved much higher accuracy and produced fewer classification errors across all categories. This suggests that the behavioral features in the dataset follow patterns that are easier to separate using a linear classification approach rather than a distance-based method. Overall, Logistic Regression was the stronger model for this prediction task and provided more reliable results for classifying social media addiction levels in this dataset.



5. Conclusion

In this project, I used machine learning classification techniques to predict social media addiction levels among Gen-Z users based on behavioral usage data. After exploring the dataset and applying preprocessing steps, I trained both Logistic Regression and K-Nearest Neighbors models to compare their performance.

The results showed that Logistic Regression performed significantly better, achieving about 99% accuracy compared to about 73% accuracy for KNN. This suggests that the behavioral features in the dataset follow patterns that are easier to separate using a linear classification approach. Overall, this project helped me apply concepts from class to a real-world dataset and better understand how preprocessing and model selection affect prediction performance.

6. References

SharmajiCoder. Gen-Z Social Media Usage Dataset. Kaggle,
<https://www.kaggle.com/datasets/sharmajicoder/gen-z-social-media-usage-dataset>

.

Pedregosa, Fabian, et al. "Scikit-learn: Machine Learning in Python." Journal of Machine Learning Research, vol. 12, 2011, pp. 2825–2830.

Python Software Foundation. Python Language Reference, Version 3. Python Software Foundation,
<https://www.python.org/>

.

McKinney, Wes. "Data Structures for Statistical Computing in Python." Proceedings of the 9th Python in Science Conference, 2010.

Hunter, J. D. "Matplotlib: A 2D Graphics Environment." Computing in Science & Engineering, vol. 9, no. 3, 2007, pp. 90–95.

7. Acknowledgments

Some online resources were used while completing this project to support understanding of machine

learning concepts and implementation steps. Python libraries such as pandas, scikit-learn, matplotlib, and seaborn were used for data preprocessing, visualization, and model training. I also used AI assistance as a reference while organizing the project structure, debugging code, and improving clarity in written explanations. All modeling decisions, dataset selection, preprocessing steps, and evaluation methods were implemented and verified by me as part of the learning process.

8. Source Code